Question 051 | Question bank page 5 | Q.1

Develop a Progressive Web Application (PWA) that displays a custom offline page when the user loses internet connectivity.

Project type: Progressive Web Application

Steps to perform

1. Create an ASP.NET Core MVC project or a suitable static web project in Visual Studio.

2. Add HTML or Razor pages, manifest.json, service-worker.js, and any referenced icons under wwwroot.

3. Register the service worker and link the manifest in the HTML layout.

4. Run the site on localhost or HTTPS and inspect it in a modern browser.

5. Use browser Developer Tools to check manifest, service worker, cache and offline operation. Push notifications require additional permission and subscription/server setup.

Closest supplied practical: Adding a line – app.UseStaticFiles(); Create a PWA that displays a custom offline page when the user is not connected to the internet.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create an MVC project named OfflinePWA. Add the home page, offline page, manifest and
service worker; register the service worker in the layout and enable static files.
Filename: index.cshtml    Location: Views/Home/Index.cshtml
<div class="text-center">
    <h1>College Information Portal</h1>
    <h3>Welcome to XYZ College</h3>
    <p>This website supports Offline Mode.</p>
</div>
Filename: offline.html    Location: wwwroot/offline.html
<!DOCTYPE html>
<html>
<head>
    <title>Offline</title>
</head>
<body>
    <h1>No Internet Connection</h1>
    <h3>Please check your internet connection.</h3>
</body>
</html>
Filename: manifest.json    Location: wwwroot/manifest.json
{
  "name": "Offline PWA",
  "short_name": "OfflinePWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
Filename: Service-Worker    Location: wwwroot/service-worker.js
const CACHE_NAME = "offline-cache-v1";
const urlsToCache = [
    "/",
    "/offline.html"
];
self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
```

```
});
self.addEventListener("fetch", event => {
    event.respondWith(
        fetch(event.request)
            .catch(() => {
                return caches.match("/offline.html");
            })
    );
});
Filename: Layout    Location: Views/Shared/_Layout.cshtml
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - OfflinePWA</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/OfflinePWA.styles.css" asp-append-version="true" />
    <link rel="manifest" href="~/manifest.json" />

</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom
box-shadow
mb-3">
            <div class="contai

ner-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">OfflinePWA</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse"
aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>
    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
    </div>
    <footer class="border-top footer text-muted">
        <div class="container">
            &copy; 2026 - OfflinePWA - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
        </div>
    </footer>
    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>
    @await RenderSectionAsync("Scripts", required: false)
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'));
        }
    </script>
</body>
</html>
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.